Inertial Navigation

Exercise 3: Strapdown Algorithm

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INTRODUCTION

Inertial Navigation Systems (INS) provide position, velocity, and attitude information by integrating measurements from inertial sensors, typically accelerometers and gyroscopes. Unlike satellite-based navigation systems, an INS operates autonomously and does not rely on external signals. This property makes inertial navigation indispensable in environments where external positioning information is unavailable or unreliable. However, the accuracy of an INS is strongly affected by sensor errors and numerical integration effects, which lead to a continuous growth of navigation errors over time (the *position* errors grow unbounded, while *velocity* errors may be bounded (horizontal) or unbounded (vertical) depending on the channel).

In this exercise, a complete strapdown inertial navigation algorithm is implemented and analyzed in the local navigation frame using the North-East-Down (NED) convention. The strapdown approach computes the navigation solution by numerically integrating the inertial measurements while continuously updating the orientation between the body frame of the IMU and the navigation frame. Correct initialization of the system, in particular the initial attitude, velocity, and position, is essential for a meaningful navigation solution [1].

The dataset used in this work was recorded with a navigation-grade fiber-optic gyroscope IMU (iMAR iIMU-FSAS) mounted on an automated platform. It consists of a static initialization phase followed by a controlled kinematic motion lasting approximately 52 seconds. During the static phase, the inertial measurements are used to estimate the local gravity vector, Earth rotation rate, and geographic latitude, which serve as inputs for the initial alignment. The initial orientation is determined using an Euler-angle-based alignment method combining accelerometer leveling and gyrocompassing.

Following initialization, the strapdown algorithm is applied to the kinematic dataset to propagate attitude, velocity, and position in the navigation frame. The resulting orientation, velocity magnitude, and horizontal trajectory are evaluated and visualized. In addition, a theoretical performance analysis is carried out using a linearized error-state model and the Van Loan discretization method to quantify the expected growth of velocity and position uncertainty due to sensor white noise [2]. Finally, the transportation rate is analyzed and its relevance for this specific experiment is discussed.

The objective of this report is to implement a basic but complete inertial navigation solution, to analyze its performance, and to relate the observed navigation errors to theoretical expectations and physical system effects.

DATA

Experimental setup and measurement system

The dataset used in this exercise was recorded during a controlled experiment conducted on the measurement roof of Leibniz University Hannover. The inertial measurements were acquired using an iMAR iIMU-FSAS navigation-grade IMU mounted on the automated platform KATE, which is additionally equipped with an optical trace sensor. The optical trace sensor provides a reference trajectory for validation purposes, while the inertial navigation solution itself is computed using IMU data only.

The IMU consists of three closed-loop fiber-optic gyroscopes and three servo accelerometers and operates at a data rate of 200 Hz. The experiment is designed to evaluate a complete strapdown inertial navigation algorithm in the local North-East-Down (NED) navigation frame under controlled conditions

Reference trajectory and motion characteristics

During the kinematic phase of the experiment, the KATE platform follows a predefined elliptic(-ish) closed trajectory on the horizontal plane. The reference track is characterized by the following geometric and kinematic parameters:

- Semi-major axis: 4.10 m
- Semi-minor axis: 1.60 m
- Corner radius: 0.80 m
- Controlled velocity: approximately 1 m/s
- Number of laps: 5

This trajectory is executed after a prolonged static initialization phase and serves as a well-defined motion pattern for evaluating the behavior of the strapdown navigation solution. The reference trajectory and its parameters are documented in the exercise discussion slides and illustrated by schematic and photographic material.

Dataset structure and measurement phases

The IMU data are provided in MATLAB format (**ex03_10075939.mat**) and are divided into two distinct measurement phases:

- Static initialization phase (approximately 648 s), during which the platform remains stationary.
- Kinematic motion phase (approximately 52 s), during which the platform moves along the elliptic reference trajectory.

Each measurement epoch contains seven columns with the structure:

$$\{t, \omega_{ib,x}^{b}, \omega_{ib,y}^{b}, \omega_{ib,z}^{b}, f_{ib,x}^{b}, f_{ib,y}^{b}, f_{ib,z}^{b}\}$$

where t denotes time in seconds, ω_{ib}^{b} are the angular rates measured by the gyroscopes in the body frame (rad/s), and f_{ib}^{b} are the specific forces measured by the accelerometers in the body frame (m/s²).

During the static phase, the accelerometer measurements are dominated by gravity and the gyroscope measurements mainly contain the Earth's rotation rate projected into the body frame. These measurements are used to estimate the local gravity magnitude, Earth rotation rate, and geographic latitude, which are required for initial alignment. The kinematic phase data are subsequently used to propagate attitude, velocity, and position using the strapdown navigation equations.

IMU noise characteristics and required additional data

For the theoretical performance analysis in Task 3, the dataset alone is not sufficient. Additional stochastic sensor parameters are required and are taken from the iMAR iIMU-FSAS datasheet [3]. The relevant white-noise density parameters are:

- Gyroscope angular random walk: 0.15 °/√h
- Accelerometer noise density: < 50 µg/√Hz

These values are converted into SI units and used to construct the continuous-time noise power spectral density matrix for accelerometer and gyroscope noise in the linearized error-state model. The noise parameters are assumed to be white and uncorrelated between axes, as specified in the datasheet IMU_FSAS.

Additional assumptions and constants

The following constants and assumptions are used throughout the analysis:

- Earth rotation rate derived from the sidereal day.
- Constant gravity magnitude estimated from static accelerometer data.
- Constant latitude during the experiment.
- Fixed Earth radius and constant ellipsoidal height for transportation-rate computation.

- Initial position and velocity assumed to be zero.
- Initial attitude uncertainty of 1 mrad per axis, used to initialize the covariance matrix for the performance analysis.
- These assumptions follow the exercise definition and the theoretical framework provided in the course material

The complete processing chain of this project, including initial alignment, strapdown mechanization, performance analysis, and visualization, is implemented in **Python**. The implementation makes use of standard scientific computing libraries such as NumPy, SciPy, and Matplotlib for numerical integration, matrix operations, and plotting.

METHOD

Part 1: Initial alignment and initialization (static data)

This part establishes the integration constants required to start the strapdown mechanization in the local navigation frame n (NED). Using the static IMU segment, local gravity, the Earth rotation rate direction, and the initial attitude are estimated. This follows Task 1 of the exercise definition.

The static data block contains epochs:

$$\{t_k, \omega_{ib}^b(t_k), f_{ib}^b(t_k)\}, \qquad k = 1, \dots, N_s \tag{1}$$

with time t_k in seconds, gyroscope angular rates $\omega_{ib}^b = [\omega_x, \omega_y, \omega_z]^T$ in rad/s, and accelerometer specific force $f_{ib}^b = [f_x, f_y, f_z]^T$ in m/s^2, all expressed in the body frame b. The sampling interval is estimated as the mean time difference:

$$\Delta t_s = \text{mean}(t_k - t_{k-1}) \tag{2}$$

(as implemented in the code using `np.mean(np.diff(Time_static))`). To reduce measurement noise during static alignment, mean values are computed:

$$\begin{aligned} \bar{f}_{ib}^b &= \frac{1}{N_s}\sum_{k=1}^{N_s} f_{ib}^b(t_k) \\ \bar{\omega}_{ib}^b &= \frac{1}{N_s}\sum_{k=1}^{N_s} \omega_{ib}^b(t_k) \end{aligned} \tag{3}$$

these averaged vectors are used as approximations of the gravity direction (from accelerometers) and Earth-rate direction (from gyroscopes) during standstill. Under static conditions and neglecting small sensor errors, the magnitude of the mean specific force approximates the local gravity magnitude:

$$g \approx ||\bar{f}^b|| \tag{4}$$

This estimate is used later as the gravity magnitude in the navigation equations (in NED: down-positive). The exercise explicitly requires estimating the local gravity vector and Earth rotation rate from the static segment.

The navigation states are initialized as required by Task 1(a):

$$x_0^n = [0,0,0]^T, \quad v_0^n = [0,0,0]^T$$

meaning the tangential-plane position and velocity are set to zero at the start epoch.

The initial attitude is computed using a Euler-angle alignment approach. Roll and pitch are obtained from the gravity direction measured by the accelerometers (leveling). Using $\bar{f}^b = \left[\bar{f}_{ib,x}^b, \bar{f}_{ib,y}^b, \bar{f}_{ib,z}^b\right]^{\boldsymbol{T}}$ the code applies:

$$\phi_0 = \operatorname{arctan2}\left(-\bar{f}_{ib,y}^b, -\bar{f}_{ib,z}^b\right) \tag{5}$$

$$\theta_0 = arctan\left(\frac{\bar{f}_{ib,x}^b}{\sqrt{\left(\bar{f}_{ib,y}^b\right)^2 + \left(\bar{f}_{ib,z}^b\right)^2}}\right)$$

Here, ϕ_0 is roll and θ_0 is pitch. The signs follow the convention used consistently in the implementation and the NED definition.

To estimate yaw, the mean gyro vector is "leveled", i.e., transformed into the intermediate navigation frame after roll and pitch compensation. With the rotation matrices $C1(\phi)$ and $C2(\theta)$,

$$\omega_{ib}^{n'} = C_2^T(\theta_0)\, C_1^T(\phi_0)\bar{\omega}_{ib}^b \tag{6}$$

The heading (yaw) is then computed as:

$$\psi_0 = \operatorname{arctan2}\left(-\omega_{ib,y}^{n'}, \omega_{ib,x}^{n'}\right) \tag{7}$$

The resulting initial direction cosine matrix (DCM) from body to navigation frame is formed using the Euler sequence defined in the course convention:

$$C_n^b(0) = C1(\phi_0)C2(\theta_0)C3(\psi_0) \tag{8}$$

This initial attitude is the orientation integration constant required by Task 1(b). The exercise additionally requires estimating the latitude from static measurements. The mean gyro vector provides the direction of the Earth rotation rate. By normalizing this vector:

$$\hat{\omega}^b = \frac{\bar{\omega}_{ib}^b}{\left\|\bar{\omega}_{ib}^b\right\|} \tag{9}$$

$$\hat{\omega}^n = \frac{C_b^n(0)\,\hat{\omega}^b}{\left\|C_b^n(0)\,\hat{\omega}^b\right\|}$$

In the NED frame, the Earth rotation direction is:

$$\hat{\omega}_{ie}^n = \begin{bmatrix} \cos(\varphi) \\ 0 \\ -sin(\varphi) \end{bmatrix} \tag{10}$$

therefore, the latitude estimate follows from the ratio of down and north components:

$$\varphi = arctan2(-\hat{\omega}_{ie,z}^{n},\ \hat{\omega}_{ie,x}^{n}) \tag{11}$$

This estimated latitude is then used in Part 2 to form the Earth rotation vector in the navigation frame.

Part 2: Strapdown mechanization using kinematic data

After completion of the static initialization and initial alignment described in Part 1, the inertial navigation solution is propagated over the kinematic measurement interval. The aim of this part is to compute the time evolution of attitude, velocity, and position in the local navigation frame using a strapdown inertial navigation algorithm, based solely on IMU measurements. This corresponds to Task 2 of the exercise.

The navigation frame is defined as the local NED frame. Due to the short duration of the experiment and the limited spatial extent of the trajectory, a local tangential-plane representation is sufficient.

Time discretization and initial conditions

The strapdown mechanization is implemented in discrete time. The sampling interval Δt\Delta tΔt is estimated from the kinematic IMU time stamps as the mean difference between consecutive epochs, as already defined in Part 1. This time step is assumed constant over the entire motion interval.

The initial navigation states at the beginning of the kinematic phase are taken directly from the static alignment results obtained in Part 1:

- the initial direction cosine matrix $C_b^n(t_0)$,
- zero initial velocity $v^n(t_0) = 0$,
- zero initial position in the tangential plane $x^n(t_0) = 0$.

These values act as integration constants for the subsequent strapdown propagation.

Earth rotation and gravity model

The inertial navigation equations require knowledge of the Earth rotation expressed in the navigation frame. Using the latitude estimate $\hat{\varphi}$^ obtained from static gyrocompassing in Part 1, the Earth rotation vector in NED coordinates is constructed as:

$$\omega_{ie}^{n} = \Omega_E \begin{bmatrix} \cos(\hat{\varphi}) \\ 0 \\ -\sin(\hat{\varphi}) \end{bmatrix}, \qquad \Omega_E = (2\pi)/T_s \tag{12}$$

where Ts=86164.09 s is the sidereal day. This formulation follows directly from the geometry of the Earth rotation axis expressed in a local-level frame.

Gravity is modeled as constant during the experiment and expressed in the NED frame as:

$$g^n = \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} \tag{13}$$

The gravity magnitude $\boldsymbol{g}$ is estimated from the static accelerometer data in Part 1. Since the platform follows a small closed trajectory with negligible change in latitude and height, variations of gravity can be safely neglected.

Attitude propagation

The attitude of the body frame with respect to the navigation frame is represented by the direction cosine matrix C_b^n. Its continuous-time evolution is governed by [3]:

$$\dot{C}_b^n = C_b^n \, [\omega_{nb}^b \times] \tag{14}$$

Where $[\cdot\times]$ denotes the skew-symmetric cross-product matrix, and ω_{nb}^b is the angular rate of the body frame with respect to the navigation frame, expressed in the body frame.

The gyroscopes measure the angular rate ω_{ib}^b of the body frame with respect to the inertial frame. To obtain ω_{nb}^b, both the Earth rotation and the rotation of the navigation frame due to motion over the Earth's surface must be removed:

The gyroscopes measure the angular rate ω_{ib}^b of the body frame with respect to the inertial frame. The Earth rotation expressed in the body frame is obtained as:

$$\omega_{nb}^b = \omega_{ib}^b - (C_b^n)^T (\omega_{ie}^n + \omega_{en}^n) \tag{15}$$

Here, ω_{en}^n is the transportation rate, which accounts for the rotation of the navigation frame caused by movement over the curved Earth surface. In accordance with the clarification received from the course instructor, the transportation rate is included directly in the strapdown mechanization.

For discrete-time implementation, the attitude update over one epoch is computed using the matrix exponential:

$$C_b^n(t_{k+1}) = C_b^n(t_k)\exp([\Delta\theta_k \times]), \qquad \Delta\theta_k = \omega_{nb}^b(t_k)\,\Delta t \tag{16}$$

This formulation is mathematically equivalent to the Rodrigues rotation formula presented in the discussion slides and is commonly used for numerical stability. To counteract numerical drift and preserve orthonormality, the direction cosine matrix is re-orthogonalized after each update using singular value decomposition:

$$C_b^n \leftarrow UV^T, \text{ with } \quad C_b^n = U\,\Sigma\,V^T \tag{17}$$

Euler angles (roll, pitch, yaw) are extracted from C_b^n for visualization purposes as required in Task 2(b).

Velocity propagation

The navigation-frame velocity is propagated using the full strapdown velocity equation [2]:

$$\dot{v}^n = C_b^n f_{ib}^b - (2\omega_{ie}^n + \omega_{en}^n) \times v^n + g^n \tag{18}$$

The first term represents the specific force measured by the accelerometers, transformed from the body frame into the navigation frame. The second term accounts for Coriolis and transport effects caused by Earth rotation and navigation-frame rotation, respectively. The last term represents the gravity acceleration.

The specific force in the navigation frame is computed as:

$$f^n = C_b^n f^b \tag{19}$$

The velocity is propagated using forward Euler integration. Given the short duration of the experiment and the high sampling rate of the IMU, this integration scheme is sufficient.

$$v^n(t_{k+1}) = v^n(t_k) + \dot{v}^n(t_k)\Delta t \tag{20}$$

Position propagation

The position in the local tangential plane is obtained by integrating the navigation-frame velocity:

$$\dot{x}^n = v^n \tag{21}$$

For improved numerical accuracy, the position update is performed using the trapezoidal rule,

$$x^n(t_{k+1}) = x^n(t_k) + 1/2\,(v^n(t_k) + v^n(t_{k+1}))\,\Delta t \tag{22}$$

This Cartesian representation is appropriate for the small spatial extent of the trajectory and avoids the need for geodetic coordinate transformations.

Following Task 2(b), the following quantities are computed and visualized:

- Roll, pitch, and yaw angles over time
- Velocity magnitude

$$||v^n|| = \sqrt{(v_N^2 + v_E^2 + v_D^2)} \tag{23}$$

- Horizontal trajectory in the tangential plane (East vs. North)

These outputs allow qualitative evaluation of the strapdown navigation solution and serve as input for the subsequent performance analysis.

Part 3: Performance analysis (linear error-state model + Van Loan)

The purpose of Part 3 is to quantify the expected growth of navigation errors caused by inertial sensor white noise during the kinematic motion. Unlike Part 2, which computes a deterministic navigation solution, this part provides a theoretical performance assessment by propagating the covariance of a linearized inertial navigation error model. The formulation follows Task 3 of the exercise and the corresponding discussion material.

Error-state definition

The inertial navigation error is modeled using a linear error-state vector composed of position, velocity, and attitude errors in the local navigation frame:

$$\delta x = \begin{bmatrix} \delta r^n \\ \delta v^n \\ \delta \psi \end{bmatrix} \in doubleR^9 \quad \delta r^n, \delta v^n, \delta \psi \in R^3 \tag{24}$$

Here, δr^ndenotes the position error in the tangential plane, δv^n the velocity error, and $\delta \psi$ the small-angle attitude error vector. This 9-state structure is explicitly defined in the Exercise 3 discussion slides and represents the standard minimal error representation for strapdown INS performance analysis.

Linearized error dynamics

Under the assumptions stated in the exercise discussion material—namely constant latitude, constant gravity, local-level (NED) navigation frame, and neglect of transport-rate effects in the performance model—the linearized continuous-time error dynamics are written as:

$$\delta \dot{x} = A\, \delta x + B\, w \tag{25}$$

with driving noise vector:

$$w = \begin{bmatrix} \delta f \\ \delta \omega \end{bmatrix} \in R^6 \tag{26}$$

represents accelerometer and gyroscope white noise. The system matrices are defined as:

$$A = \begin{bmatrix} 0_3 & \mathrm{I}_3 & 0_3 \\ 0_3 & -2[\omega_{ie}^n \times] & -[f^n \times] \\ 0_3 & 0_3 & 0_3 \end{bmatrix}, \quad B = \begin{bmatrix} 0_3 & 0_3 \\ \mathrm{I}_3 & 0_3 \\ 0_3 & \mathrm{I}_3 \end{bmatrix} \tag{27}$$

This structure reflects the physical coupling mechanisms in a strapdown INS:

- the kinematic relationship between position and velocity errors,
- the Coriolis coupling between velocity errors and Earth rotation,
- and the coupling of attitude errors into velocity errors through the specific force.
- The operator$[\cdot\times]$ denotes the skew-symmetric matrix such that $[a \times]b = a \times b$.

Earth rotation in the navigation frame

Using the latitude estimate $\hat{\varphi}$ obtained in Part 1, the Earth rotation vector expressed in the NED navigation frame is given by [4]:

$$\omega_{ie}^n = \Omega_E \begin{bmatrix} \cos(\hat{\varphi}) \\ 0 \\ -sin(\hat{\varphi}) \end{bmatrix}, \quad \Omega_E = \frac{2\pi}{T_s}, \qquad T_s = 86164.09 \text{ s} \tag{28}$$

This expression follows directly from the geometry of the Earth rotation axis expressed in a local-level frame and is used consistently in both the mechanization (Part 2) and the performance model.

Gravity versus specific force (critical sign convention)

A frequent source of error in INS performance modeling is confusion between gravity acceleration and specific force. In the NED frame, the gravity acceleration vector is as in Eq (13), since the Down axis is positive. However, the linearized error dynamics do **not** depend on gravity directly, but on the **specific force** f, defined as: $f = a - g$

For the performance analysis, a static or gravity-only condition is assumed, such that the true acceleration $a \approx 0$. Consequently, the specific force in the navigation frame becomes:

$$f^n \approx -g^n = \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} \tag{29}$$

This vector appears in the coupling term $-[f^n \times]\delta\psi$ of the $\boldsymbol{A}$ matrix and explains why the sign differs from the gravity term used in the mechanization equations of Part 2.

Sensor white-noise model

The inertial sensor noise is modeled as zero-mean, white, and uncorrelated between axes. The continuous-time power spectral density (PSD) matrix is therefore chosen as diagonal:

$$Q_u = \mathrm{diag}(\sigma_f^2, \sigma_f^2, \sigma_f^2,\ \sigma_\omega^2, \sigma_\omega^2, \sigma_\omega^2) \tag{30}$$

The noise parameters are taken from the iMAR iIMU-FSAS datasheet:

- Gyro angular random walk: $0.15\ °/h$
- Accelerometer noise density: $50\ \mu g/Hz$

They are converted to SI units (as in your code):

$$\sigma_\omega = (\frac{0.15}{60})\frac{\pi}{180}[rad/s] \tag{31}$$
$$\sigma_f = 50 \times 10^{-6}\, g_0[m/s2/Hz], \qquad \mathrm{g0} = 9.80665\ m/s^2$$

Van Loan discretization

To propagate the covariance matrix in discrete time with step size Δt, the continuous-time model is discretized using the Van Loan method. The following augmented matrix is formed [5]:

$$M\ = \begin{bmatrix} -A\Delta t & B\ Q_u B^T \Delta t \\ 0 & A^T \Delta t \end{bmatrix} \tag{32}$$

and its matrix exponential is computed:

$$N = \exp(\mathrm{M}) \tag{33}$$

From the block structure of N, the discrete-time state transition matrix and process noise covariance are obtained as:

$$\Phi = N_{22}^T, \qquad Q_k = \Phi N_{12} \tag{34}$$

This procedure ensures a consistent discretization of both the system dynamics and the driving noise.

Covariance propagation and performance metrics

The error covariance is propagated recursively according to:

$$P_{k+1} = \Phi P_k \Phi^T + Q_k \tag{35}$$

Initial position and velocity errors are assumed zero, while the initial attitude uncertainty is set to: $\sigma_{\psi,0} = 1\ mrad$, which result in:

$$P_0 = \mathrm{diag}\left(0,0,0,0,0,0,\sigma_{\psi,0}^2,\sigma_{\psi,0}^2,\sigma_{\psi,0}^2\right) \tag{36}$$

As required by Task 3, the performance metrics are the magnitudes of the position and velocity standard deviation vectors:

$$\begin{aligned} \left||\sigma_r|\right| &= \sqrt{P_{11} + P_{22} + P_{33}} \\ \left||\sigma_v|\right| &= \sqrt{P_{44} + P_{55} + P_{66}} \end{aligned} \tag{37}$$

which are evaluated over the 52-second motion interval and plotted to assess the expected error growth.

Part 4: Transportation rate of the navigation frame and relevance assessment

The purpose of Part 4 is to analyze the transportation rate of the navigation frame and to assess its relevance for the given experiment. While the transportation rate is included in the strapdown mechanization in Part 2, its physical impact depends on the platform velocity, the trajectory size, and the Earth model used.

Physical meaning of the transportation rate

The transportation rate ω_{en}^ndescribes the angular velocity of the local navigation frame with respect to the Earth-fixed frame caused by motion over the Earth's surface. Unlike the Earth rotation rate, which depends only on latitude, the transportation rate is proportional to the platform velocity and inversely proportional to the Earth radius.

For small-scale motion and short experiment duration, the Earth can be approximated as a sphere, and the navigation frame can be treated as moving over a locally flat surface.

Transportation rate model used in this experiment

Under a spherical Earth approximation with constant radius R and constant height h, the transportation rate in the NED frame is expressed as:

$$\omega_{en}^{n} = \begin{bmatrix} \omega_{en,N}^{n} \\ \omega_{en,E}^{n} \\ \omega_{en,D}^{n} \end{bmatrix} = \begin{bmatrix} \dfrac{v_E}{R_E + h} \\ -\dfrac{v_N}{R_N + h} \\ -\dfrac{v_E tan\varphi}{R_E + h} \end{bmatrix} \tag{38}$$

where v_N and v_E denote the north and east velocity components obtained from the strapdown mechanization in Part 2. and φ is the latitude estimate from Part 1. In this exercise, a spherical Earth radius of $R = 6378000$ m and a constant height $h = 70$ m is used. These approximations are justified by the small spatial extent of the KATE trajectory and the short experiment duration.

Using the velocity history from Part 2, the transportation rate components are computed at each epoch. To evaluate its overall magnitude, the Euclidean norm:

$$||\omega_{en}^{n}|| = \sqrt{\left(\omega_{en,N}^{n}\right)^2 + \left(\omega_{en,E}^{n}\right)^2 + \left(\omega_{en,D}^{n}\right)^2} \tag{39}$$

is computed and compared to the magnitude of the Earth rotation rate

$$||\omega_{ie}^{n}|| = \Omega_E = \frac{2\pi}{T_s}, \qquad T_s = 86164.09 \text{ s} \tag{40}$$

This comparison allows an order-of-magnitude assessment of the transportation-rate effect relative to Earth rotation.

Relevance for the present experiment

The KATE platform follows a closed elliptic(-like) trajectory with an approximate velocity of 1 m/s and a spatial extent of only a few meters. Under these conditions, the transportation rate is several orders of magnitude smaller than the Earth rotation rate. Consequently, although it is correctly included in the mechanization, its contribution to the overall navigation solution remains very small over the 52-second experiment duration.

This analysis confirms that the use of a spherical Earth model and constant gravity is sufficient for Exercise 3 and explains why transportation-rate effects have only a minor influence on the final navigation results.

RESULT AND DISCUSSION

Part 1: Initialization Using Static IMU Data

Using the static IMU segment, the mean specific force and angular rate vectors in the body frame were computed by time-averaging the measurements. These mean values form the basis for the initialization of the inertial navigation system.

The resulting static mean vectors are:

$$\bar{f}_b = \begin{bmatrix} -0.03296 \\ -0.13416 \\ -9.82606 \end{bmatrix} \left[\frac{\mathrm{m}}{\mathrm{s}^2}\right], \qquad \bar{\omega}_b = \begin{bmatrix} -2.62 \\ -3.19 \\ -5.19 \end{bmatrix} \times 10^{-5} \left[\frac{\mathrm{rad}}{\mathrm{s}}\right]$$

From these measurements, the local gravity magnitude and the Earth rotation rate magnitude were estimated according to the initialization equations described in the Methods section. The following values were obtained:

- Estimated local gravity: $g = 9.827027 \left[\frac{\mathrm{m}}{\mathrm{s}^2}\right]$

Estimated Earth rotation magnitude: $| \omega_{ie} | = 13.685258 \left[\frac{\mathrm{deg}}{\mathrm{h}}\right]$

The gravity estimate is close to the expected local gravitational acceleration. The slight deviation from the nominal value can be attributed to accelerometer bias and scale-factor effects. The estimated Earth rotation rate is of the correct order of magnitude; the underestimation relative to the theoretical value is consistent with gyro bias, noise, and the finite averaging interval.

The initial position and velocity in the navigation frame were set to zero, as required for the start of the strapdown integration.

Based on the static mean measurements, the initial roll, pitch, and yaw angles were determined using accelerometer leveling and gyrocompassing, following the procedure outlined in the Methods section. The resulting initial Euler angles are:

$$[\phi_0,\ \theta_0,\ \psi_0]^T = [-0.7822^\circ,\ -0.1922^\circ,\ -49.8020^\circ]^T$$

From these angles, the initial direction cosine matrix describing the transformation from body frame to navigation frame was constructed. The resulting matrix is:

$$C_n^b(0) = \begin{bmatrix} -0.6454278 & 0.7637765 & 0.0082632 \\ -0.7638139 & 0.6453364 & 0.0113734 \\ 0.0033541 & -0.0136522 & 0.9999012 \end{bmatrix}$$

The matrix is orthonormal and represents a small roll and pitch offset combined with a heading of approximately −50°, which is consistent with the experimental setup.

Using the aligned Earth rotation vector in the navigation frame, the latitude was estimated as:

$$\hat{\varphi} = 51.966915^\circ$$

This value is in very good agreement with the true latitude of the experiment location (approximately 52°), indicating that the gyrocompassing-based yaw determination was successful.

The results of Part 1 demonstrate that the initialization and initial alignment were performed correctly. The estimated gravity and Earth rotation values are physically plausible and consistent with expectations for static IMU measurements. The close agreement between the estimated and true latitude confirms that the attitude alignment provides a reliable initial orientation. These

results establish a consistent and well-conditioned set of initial integration constants for the subsequent strapdown navigation solution.

Part 2: Strapdown Navigation (Moving Data)

Based on the initial conditions obtained in Part 1, the strapdown mechanization described in the Methods section was applied to the moving IMU data. The resulting attitude, velocity, and position estimates are discussed below with reference to the corresponding figures.

According to Figure 1, which shows the evolution of roll, pitch, and yaw angles over time, the attitude solution remains stable throughout the experiment. The roll and pitch angles stay close to zero, with small oscillations on the order of one degree. This behavior indicates that the system remains well leveled and that no significant attitude divergence occurs during the integration. The observed high-frequency variations are consistent with sensor noise and small vibrations of the cart and are therefore expected in a real measurement scenario.

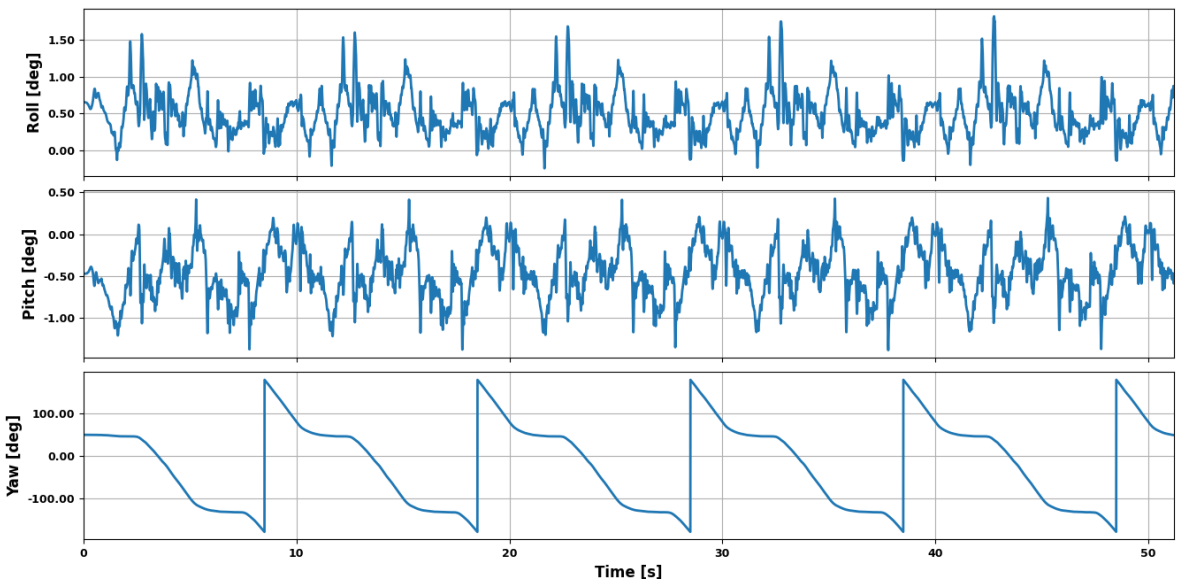

Figure 1: *Time evolution of the estimated Euler angles (roll, pitch, and yaw) obtained from the strapdown mechanization.*

The yaw angle exhibits a smooth, periodic evolution that corresponds to the repeated turning of the cart along the course. The abrupt jumps visible in the yaw plot are not caused by physical rotations of the system but are a consequence of angle wrapping at ±180°, which arises from the Euler angle representation used to extract yaw from the direction cosine matrix. Overall, the yaw behavior demonstrates that the heading propagation is consistent with the motion and that the initial alignment from Part 1 provides a reliable starting orientation.

The final attitude at the end of the experiment is:

Roll = 0.7937°, Pitch = −0.5821°, Yaw = 49.1196°.

The velocity magnitude over time is shown in Figure 2. After an initial acceleration phase, the velocity settles into a range between approximately 0.6 m/s and 1.3 m/s. This range is realistic for the cart experiment and confirms that the velocity update described in the Methods section behaves physically. The recurring increase and decrease in speed correspond to straight segments and turning maneuvers of the cart, respectively.

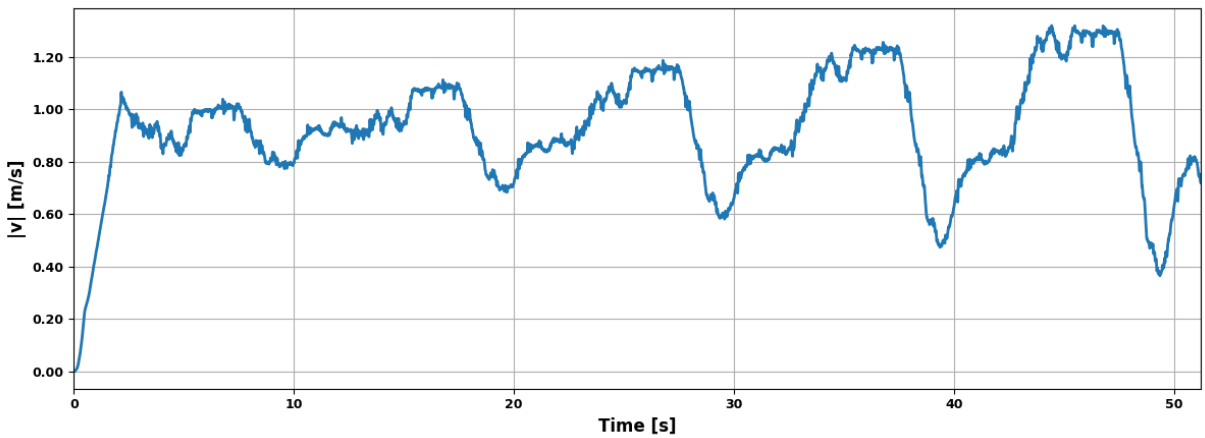

Figure 2: *Magnitude of the velocity vector in the navigation frame over time.*

The smooth evolution of the velocity magnitude, without unbounded growth or abrupt numerical artifacts, indicates that gravity compensation, Coriolis effects, and transport rate contributions are handled consistently within the mechanization. The final velocity vector at the end of the segment is:

$$v^n = \begin{bmatrix} 0.7079 \\ -0.1272 \\ 0.0326 \end{bmatrix} [\frac{m}{s}]$$

The small down component is attributed to residual attitude errors and sensor imperfections rather than actual vertical motion.

According to Figure 3, the reconstructed horizontal trajectory in the East–North plane consists of a sequence of loop-like curves that gradually drift away from the initial position. This characteristic shape is a direct consequence of the elliptic(-like) closed motion executed by the cart during the experiment, combined with the nature of unaided inertial navigation.

The repeated loop structure reflects the controlled circular and turning motion of the platform along the predefined course. Each loop corresponds to one lap of the cart, during which the heading continuously changes while the velocity magnitude remains approximately constant. As a result, the integrated horizontal velocity produces curved segments rather than straight lines, leading to the observed loop-like trajectory.

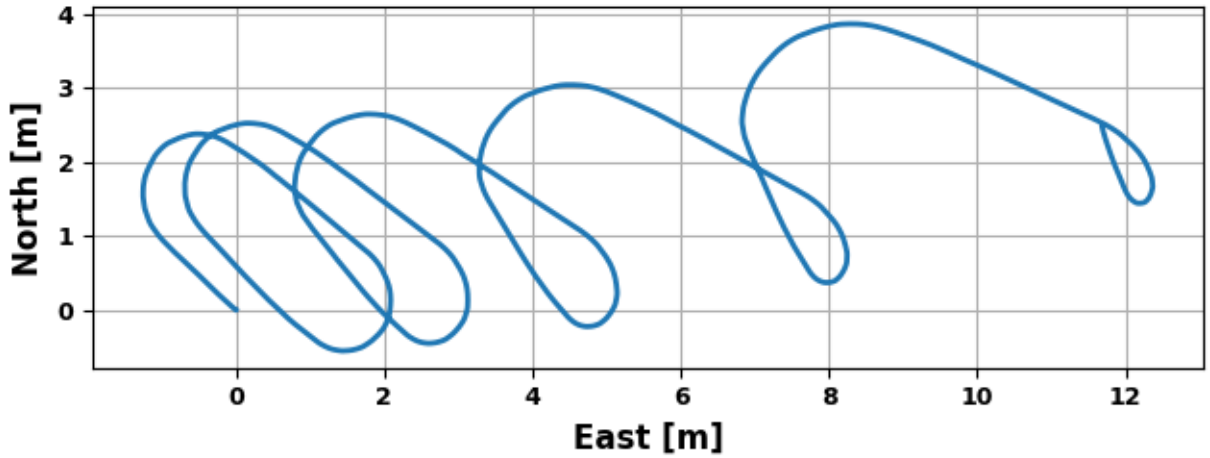

Figure 3: *Reconstructed horizontal trajectory in the local tangential plane (East vs North) obtained from inertial integration.*

The fact that the loops do not close perfectly and that the trajectory exhibits a gradual drift, most notably in the East direction, is caused by the accumulation of inertial navigation errors. Small residual attitude errors, accelerometer biases, and gyro noise are continuously integrated over time. Even very small errors in heading or horizontal acceleration lead to systematic position drift when no external corrections are applied. This behavior is typical for an unaided strapdown INS and is expected even over the relatively short duration of approximately 52 seconds.

Therefore, the overall shape of Figure 3 results from the superposition of the true closed motion pattern and the unbounded error growth inherent to inertial integration, rather than from numerical instability or implementation errors. The observed trajectory is physically plausible and consistent with the theoretical limitations of inertial navigation described in the course material

The final position estimate at the end of the experiment is:

$$x^n = \begin{bmatrix} 2.4699 \\ 11.6843 \\ 0.4428 \end{bmatrix} \text{[m]}$$

This position should not be interpreted as an absolute reference but rather as the result of accumulated inertial errors starting from the assumed initial position.

Taken together, Figures 1–3 demonstrate that the strapdown mechanization produces a physically consistent navigation solution. The attitude remains stable, the velocity magnitude is realistic, and the reconstructed trajectory reflects the expected motion pattern while exhibiting the characteristic drift of an unaided INS. These results confirm that the implementation

described in the Methods section is correct and provides a suitable basis for the subsequent error analysis and discussion in the following tasks.

Part 3: Theoretical Error Propagation

In Part 3, the theoretical growth of navigation errors was evaluated using covariance propagation based on the linearized error model described in the Methods section. The analysis focuses on the influence of stochastic IMU noise on velocity and position uncertainty and provides a theoretical reference for comparison with the deterministic strapdown solution obtained in Part 2.

The stochastic model was parameterized using the IMU noise specifications, resulting in an equivalent accelerometer noise density of

$\sigma_f = 4.90 \times 10^{-4}\ \mathrm{m/s2/Hz}$

and a gyroscope noise density of

$\sigma_\omega = 4.36 \times 10^{-5}\ \mathrm{rad/s}$

These parameters were assumed constant over the entire experiment duration.

The evolution of the velocity standard deviation magnitude is shown in Figure 4**.** According to the figure, the velocity uncertainty increases almost linearly with time, starting from zero and reaching a value of approximately $\| \sigma_v(52\,s) \| = 0.735\ \mathrm{m/s}$ at the end of the experiment.

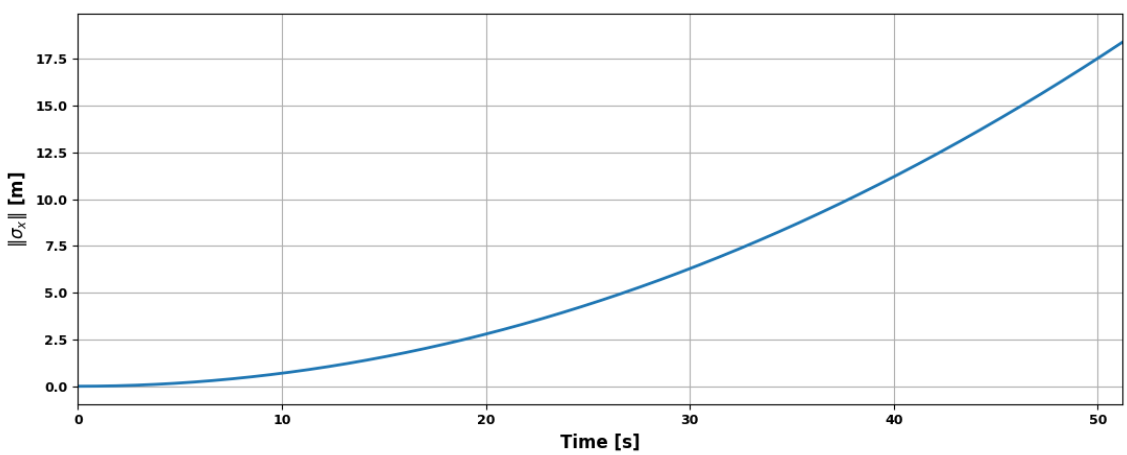

Figure 4: *Magnitude of the predicted velocity standard deviation over time obtained from covariance propagation using the stochastic IMU error model.*

This near-linear growth is consistent with the theoretical model, as white accelerometer noise integrates directly into velocity uncertainty. The smooth shape of the curve reflects the fact that only stochastic noise was considered, without bias terms or time-correlated errors. No oscillations or discontinuities are visible, which indicates that the covariance propagation is numerically stable and dominated by the assumed noise characteristics.

The predicted position uncertainty magnitude is shown in Figure 5. In contrast to the velocity uncertainty, the position standard deviation exhibits a clearly nonlinear, approximately quadratic growth over time. At the end of the experiment, the position uncertainty reaches $\| \sigma_x(52\,s) \| = 18.98$ m.

This behavior is expected, since position uncertainty results from the time integration of velocity uncertainty. As the velocity variance increases linearly, its integral leads to a faster-than-linear growth of position variance. The smooth curvature of the position uncertainty curve again reflects the simplified stochastic model and the absence of bias-driven or environmental error sources.

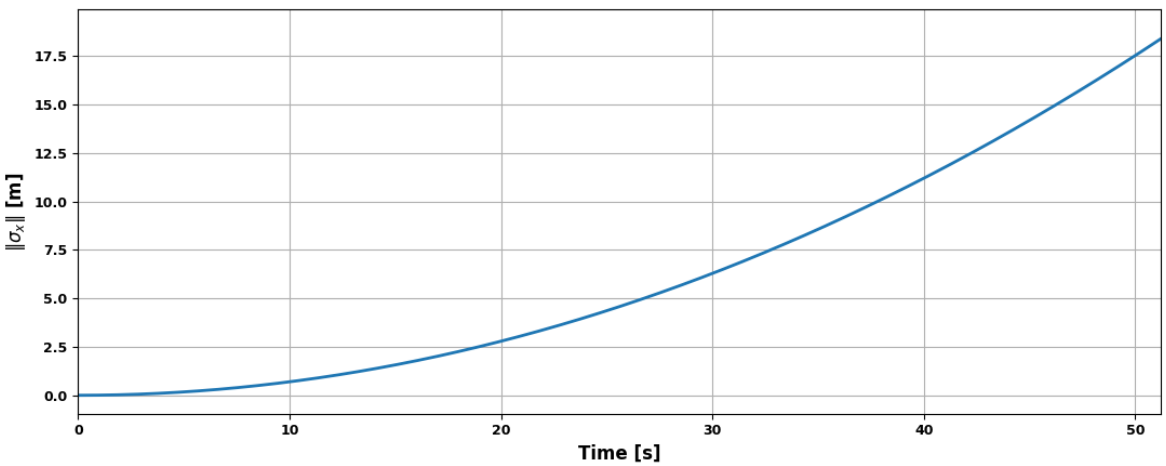

Figure 5: *Magnitude of the predicted position standard deviation over time obtained from covariance propagation.*

The theoretical uncertainties obtained in Part 3 can be compared with the final results of the strapdown solution from Part 2. At $t = 52$s the strapdown solution shows a final velocity magnitude of about 0.72 m/s and a horizontal position error of roughly 12 m. The theoretical covariance model predicts a velocity standard deviation of about 0.73 m/s and a position standard deviation of about 19 m at the same time.

The velocity error observed in the strapdown solution is therefore very close to the predicted 1σ value, while the position error remains below the predicted position uncertainty. This is expected, since the covariance model describes the typical error level over many possible realizations, whereas the strapdown result represents only a single realization, which may lie above or below the 1σ bound.

It should be noted that the theoretical model in Part 3 only considers white sensor noise. Other error sources, such as sensor biases, scale factor errors, sensor misalignment, and vibrations, were not included. These effects can cause faster error growth in practice, especially in position. Nevertheless, the theoretical results provide a useful reference and explain why noticeable position drift appears even over a short time interval, illustrating the fundamental limitations of unaided inertial navigation.

At least two important error sources were not considered in the theoretical assessment of Part 3. First, sensor biases in both accelerometers and gyroscopes were neglected. Constant or slowly varying biases introduce systematic errors that integrate over time, leading to linear growth in velocity error and quadratic growth in position error, which can dominate the error budget even over short durations. Second, sensor scale factor and misalignment errors were not modeled. Imperfect scale factors cause proportional errors in measured specific force and angular rate, while axis misalignment couples motion between axes, leading to incorrect attitude estimates and additional spurious accelerations. Both effects result in significantly faster and more structured error growth than predicted by the white-noise-only covariance model used in Part 3. Consequently, the theoretical uncertainties presented represent a lower bound on the true navigation error and explain why real strapdown solutions typically drift faster than predicted by simplified stochastic models.

In terrestrial applications with free inertial navigation over long periods (> 1 hour), the system will accumulate significant position errors that grow rapidly (often cubically with time). Regarding the specific behavior of velocity errors under a constant acceleration bias: horizontal velocity errors do not grow unbounded, whereas vertical velocity errors do grow unbounded.

Reasoning

The difference in behavior is due to the physics of the Earth's gravity field. The horizontal channel is inherently stable due to the Schuler effect: as the system drifts over the curved Earth, the gravity vector "righting" itself acts as a restoring force, causing horizontal velocity errors to oscillate with a period of roughly 84.4 minutes rather than growing to infinity. In contrast, the vertical channel is inherently unstable. A small error in height leads to an incorrect gravity calculation, which creates a positive feedback loop (e.g., thinking you are higher makes you subtract less gravity, which makes you think you are accelerating upward even faster). This causes the vertical velocity error to diverge exponentially.

A primary IMU-only technique used to reset velocity drift is the Zero-Velocity Update (ZUPT).

Reasoning

This technique exploits the known kinematic state of the vehicle or sensor. When the system detects that it is stationary—such as a car stopping at a red light or a pedestrian’s foot contacting the ground—the navigation algorithm forces the velocity state to exactly zero. This effectively "clamps" the velocity error that has accumulated during the movement phase, preventing that specific chunk of velocity drift from being integrated further into position error.

Part 4: Transportation Rate

In Part 4, the influence of the transportation rate on the navigation solution was evaluated. Although the transportation rate was already included in the strapdown mechanization in Part 2, this part aims to quantify its magnitude and assess its relevance for the given experimental scenario.

Transportation rate components

The time evolution of the three components of the transportation rate expressed in the navigation frame is shown in Figure 6. According to the figure, the North, East, and Down components of ω_{en}^{n} vary periodically over time, following the velocity pattern of the cart motion. The changes in sign and magnitude are consistent with the repeated turns and direction changes along the closed course.

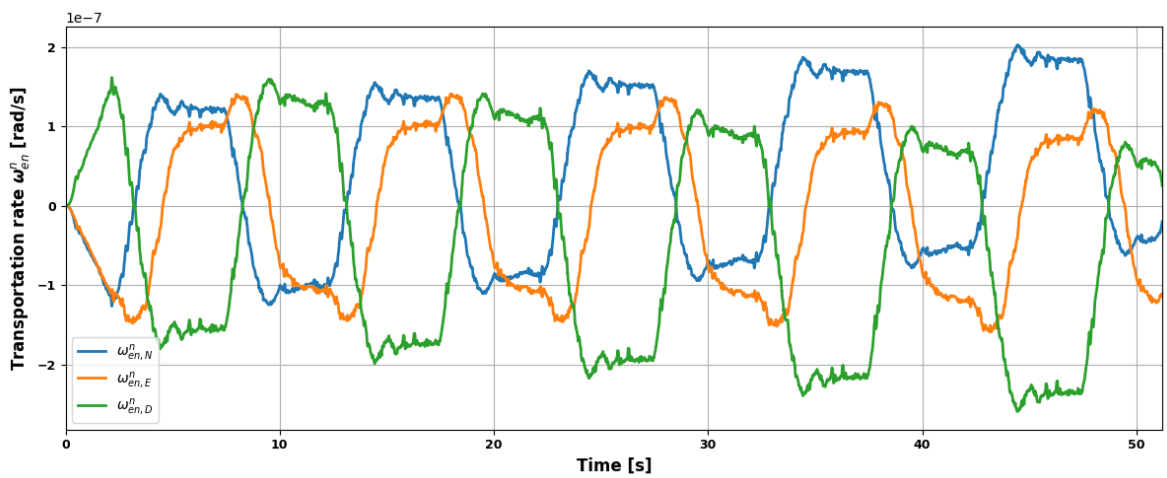

Figure 6: *Time evolution of the three components of the transportation rate expressed in the navigation frame.*

Despite this structured behavior, the absolute magnitude of the transportation rate components remains very small. The peak values are on the order of 10^{-7} rad/s, which is several orders of magnitude below typical inertial angular rates encountered in the attitude propagation.

A direct comparison between the magnitude of the transportation rate and the Earth rotation rate is shown in Figure 7. The figure clearly illustrates that the transportation rate magnitude $\|\omega_{en}^{n}\|$ is nearly constant at a very low level, while the Earth rotation rate $\|\omega_{ie}^{n}\|$ appears as a horizontal reference line at a much higher value.

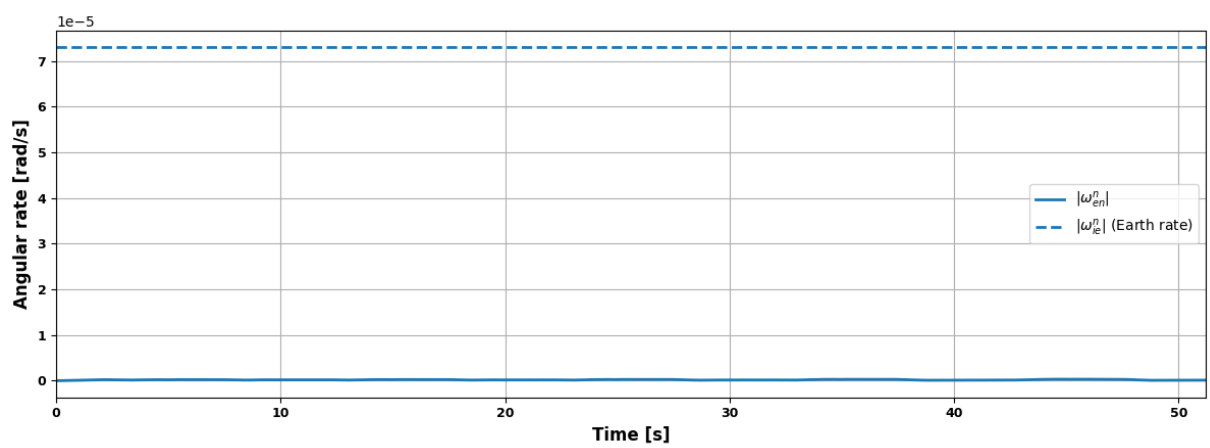

Figure 7: *Comparison of the magnitude of the transportation rate with the Earth rotation rate.*

Quantitatively, the following statistics were obtained over the experiment duration:

Mean transportation rate magnitude:

$\| \overline{\omega_{en}} \| = 1.96 \times 10^{-7}$ rad/s

Maximum transportation rate magnitude:

$max \| \omega_{en} \| = 3.31 \times 10 - 7$ rad/s

Earth rotation magnitude:

$\| \omega_{ie}^{n} \| = 7.29 \times 10^{-5}$ rad/s

These values show that the transportation rate is approximately two to three orders of magnitude smaller than the Earth rotation rate for this experiment.

The results confirm that, for the given cart motion with velocities on the order of 1 m/s and a duration of about 52 s, the transportation rate has only a minor influence on the navigation solution. While its inclusion in the mechanization is formally correct and required, its practical impact on attitude, velocity, and position is small compared to that of Earth rotation and sensor noise.

Regarding the sensitivity of the transportation rate, the dominant parameter is the vehicle velocity (v). As shown in the definition eq (38), the magnitude of the transportation rate is directly proportional to the horizontal velocity and inversely proportional to the Earth's radius (R). Since the Earth's radius is very large and effectively constant, and the latitude (φ) changes only slowly for typical terrestrial applications, variations in the transportation rate are driven almost exclusively by changes in vehicle speed. Therefore, velocity is the parameter with the biggest impact, scaling linearly with the magnitude of the transportation rate.

A realistic example where the transportation rate becomes larger than the Earth rotation rate ($\approx 15^{\circ}/h$) is a supersonic aircraft or a hypersonic missile.

Reasoning: To exceed the Earth rotation rate, the term v/R must be greater than ω_e.

Calculation: $\frac{v}{R} > \Omega_E \Rightarrow v > R \cdot \Omega_E \approx 6371\ \mathrm{km} \cdot 7.29 \times 10^{-5}\ \mathrm{rad/s} \approx 464\ \mathrm{m/s}$

Conclusion: Any vehicle traveling faster than approximately 464 m/s (Mach 1.4) near the equator will experience a transportation rate larger than the Earth's rotation rate.

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